Evaluation runs: **bank** (bank-grove)

generated 2026-11-05 from brain-data bb4df53ad 2026-11-05 15:31:21 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

No run of the stateless engine on this benchmark.

stateful

No run of the stateful engine on this benchmark.

offline

offline: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	6	366	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	30	43383	conforms	2.0	18	10 min, 16 GB	0.0	2.0	0.0	in-memory	brain05, 2026-11-02
3	140	2752874	conforms	117.9	61	10 min, 16 GB	0.0	117.8	0.1	in-memory	brain05, 2026-11-02

offline: pair **unordered**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	6	258	9	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	30	11072	157	violation	0.6	14	10 min, 16 GB	0.0	0.6	0.0	in-memory	brain05, 2026-11-02
3	140	247764	1586	violation	7.8	40	10 min, 16 GB	0.0	7.7	0.1	in-memory	brain05, 2026-11-02
4	630	4869014	14510	violation	227.0	238	10 min, 16 GB	0.1	225.6	1.4	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): 5 unordered fifo.

skipped: a smaller size had reached its limit: 4 ordered fifo, 5 unordered fifo, 5 ordered fifo, 6 unordered fifo, 6 ordered fifo, 7 ordered fifo, 7 unordered fifo, 8 unordered fifo, 8 ordered fifo, 9 ordered fifo, 9 unordered fifo.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	6	366	conforms	3.1	165	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	30	43383	conforms	3.1	160	10 min, 16 GB	3.1	2.0	0.0	in-memory	brain01, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	140	2752874	conforms	117.8	156	10 min, 16 GB	3.6	117.8	0.1	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **unordered**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	6	258	9	violation	3.1	165	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	30	11072	157	violation	3.1	160	10 min, 16 GB	3.1	0.6	0.0	in-memory	brain01, 2026-11-02
3	140	247764	1586	violation	7.8	156	10 min, 16 GB	3.6	7.7	0.1	in-memory	brain01, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	6	366	conforms	10.2	342	1 h, 250 GB		brain05, 2026-11-02
2	30		🕓 (1 h)			1 h, 250 GB		brain05, 2026-11-02
3	140		🕓 (1 h)			1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **unordered**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	6	258	110	violation	5.0	322	1 h, 250 GB	349	brain05, 2026-11-02
2	30	11072	5598	violation	645.6	973	1 h, 250 GB	820	brain05, 2026-11-02
3	140			🕓 (1 h)			1 h, 250 GB		brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 1 fifo unordered, 1 fifo ordered, 2 fifo unordered, 2 fifo ordered, 3 fifo ordered, 3 fifo unordered.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	6	366	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02
2	30	43383	conforms	2.0	10	1 h, 250 GB		brain01, 2026-11-02
3	140	2752874	conforms	109.1	63	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **unordered**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	6	258	110	violation	0.1	4	1 h, 250 GB	2	brain01, 2026-11-02
2	30	11072	5598	violation	0.5	10	1 h, 250 GB	2	brain01, 2026-11-02
3	140	247764	122182	violation	8.3	30	1 h, 250 GB	2	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	366	366	conforms	9.9	958	10 min, 16 GB	9.9	0.1	0.0	in-memory	brain01, 2026-11-02
2	43383	43383	conforms	1504.8	15241	10 min, 16 GB	1504.7	2.0	0.0	in-memory	brain01, 2026-11-02
3	2752874	2752874	conforms	20843.1	80188	48 h, 250 GB	20843.0	124.8	0.1	merge	brain01, 2026-11-04

twins offline, Rust implementation against its Java twin: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	258	258	conforms	6.4	329	10 min, 16 GB	6.4	0.1	0.0	in-memory	brain01, 2026-11-02
2	11072	11072	conforms	317.2	14570	10 min, 16 GB	317.1	0.6	0.0	in-memory	brain01, 2026-11-02
3	247764	247764	conforms	1972.0	19798	48 h, 250 GB	1971.8	9.6	0.2	merge	brain01, 2026-11-04

Not run (twins offline, Rust implementation against its Java twin). left out by the table rules (ok): 3 fifo unordered.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	366	366	conforms	9.9	958	10 min, 16 GB	0.1	9.9	0.0	in-memory	brain01, 2026-11-02
2	43383	43383	conforms	1504.8	15241	10 min, 16 GB	2.0	1504.7	0.0	in-memory	brain01, 2026-11-02
3	2752874	2752874	conforms	20843.1	80188	48 h, 250 GB	124.8	20843.0	0.1	merge	brain01, 2026-11-04

twins offline, Java implementation against its Rust twin: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	258	258	conforms	6.4	329	10 min, 16 GB	0.1	6.4	0.0	in-memory	brain01, 2026-11-02
2	11072	11072	conforms	317.2	14570	10 min, 16 GB	0.6	317.1	0.0	in-memory	brain01, 2026-11-02
3	2752874	247764	violation*	1972.0	19798	48 h, 250 GB	124.8	1971.8	0.1	merge	brain01, 2026-11-04

Not run (twins offline, Java implementation against its Rust twin). left out by the table rules (ok): 3 fifo unordered.

twins online, Rust implementation (TraceForge) against the Java twin's export

twins online, Rust implementation (TraceForge) against the Java twin's export: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3	2752874	2752874	conforms	133.8	77	48 h, 250 GB		brain01, 2026-11-04

twins online, Rust implementation (TraceForge) against the Java twin's export: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3	247764	247764	conforms	10.1	73	48 h, 250 GB		brain01, 2026-11-04

Not run (twins online, Rust implementation (TraceForge) against the Java twin's export). planned but never scheduled (grid points beyond the ladder): 1 fifo unordered, 1 fifo ordered, 2 fifo ordered, 2 fifo unordered.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	366	366	conforms	9.0	887	10 min, 16 GB	9.0	0.4	0.0	in-memory	brain02, 2026-11-03
2	43383	43383	conforms	1288.6	15323	10 min, 16 GB	1288.6	2.3	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	258	258	conforms	7.9	754	10 min, 16 GB	7.8	0.2	0.1	in-memory	brain02, 2026-11-03
2	11072	11072	conforms	240.5	14903	10 min, 16 GB	240.4	0.7	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). refused (an input missing at run time): 3 fifo unordered, 3 fifo ordered.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	366	366	conforms	9.0	887	10 min, 16 GB	0.4	9.0	0.0	in-memory	brain02, 2026-11-03
2	43383	43383	conforms	1288.6	15323	10 min, 16 GB	2.3	1288.6	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	258	258	conforms	7.9	754	10 min, 16 GB	0.2	7.8	0.1	in-memory	brain02, 2026-11-03
2	11072	11072	conforms	240.5	14903	10 min, 16 GB	0.7	240.4	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 3 fifo unordered, 3 fifo ordered.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	366	366	conforms	10.2	952	1 h, 250 GB		brain02, 2026-11-03
2	43383	43383	conforms	1305.8	15666	1 h, 250 GB		brain02, 2026-11-03
3	2752874		(1 h)			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	258	258	conforms	7.0	708	1 h, 250 GB		brain02, 2026-11-03
2	11072	11072	conforms	263.4	14822	1 h, 250 GB		brain02, 2026-11-03
3	247764		(1 h)			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **ordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	366	366	conforms	3.8	2	1 h, 250 GB		brain02, 2026-11-03
2	43383	43383	conforms	2.2	16	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **unordered**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	258	258	conforms	0.1	4	1 h, 250 GB		brain02, 2026-11-03
2	11072	11072	conforms	0.6	14	1 h, 250 GB		brain02, 2026-11-03

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). skipped: a smaller size had reached its limit: 3 fifo unordered, 3 fifo ordered.